Description

Image



Caption

Low alloy steel wrench. © ANSYS, Inc.

The material

Pure iron is soft stuff. Add carbon and heat-treat it right, and you can get a material that is almost as hard and brittle as glass, or as ductile and tough as boiler plate. 'Heat treat' means heating the steel to about 800 C to dissolve the carbon, then quenching (rapid cooling, often by dropping into cold water) and tempering - reheating it to a lower temperature and holding it there. Quenching turns the steel into hard, brittle 'martensite'; tempering slowly restores the toughness and brings the hardness down. Control of tempering time and temperature gives control of properties. It's wonderful what 1% of carbon can do. But (the inevitable 'but') the cooling rate in that initial quench has to be fast - more than 200 C/second for plain carbon steels. There is no difficulty in transforming the surface of a component to martensite, but the interior cools more slowly because heat has to be conducted out. If the component is more than a few millimeters thick, there is a problem - the inside doesn't cool fast enough. The problem is overcome by alloying. Add a little manganese (Mn), nickel (Ni), molybdenum (Mo), or chromium (Cr), and the critical cooling rate comes down, allowing thick sections to be hardened and then tempered. Adding some vanadium, V, as well allows a dispersion of carbides giving strength while retaining toughness and ductility. Chrome-molybdenum steels such as AISI4140 are used for aircraft tubing and other high strength parts. Chrome-vanadium steels are used for crank and propeller shafts and high quality tools. Steels alloyed for this purpose are called low alloy steels, and the property they have is called 'hardenability'.

Compositional summary

Fe/<1.0 C/<2.5 Cr/<2.5 Ni/<2.5 Mo/<2.5 V

General properties

Density	7.8e3			kg/m^3
Price	* 0.964	-	1.09	GBP/kg
Date first used	1930			

Mechanical properties

Young's modulus	200	-	210	GPa

Shear modulus		77	-	85	GPa
Bulk modulus		160	-	180	GPa
Poisson's ratio		0.28	-	0.29	
Yield strength (elastic limit)		469	-	1.6e3	MPa
Tensile strength		699	-	1.8e3	MPa
Compressive strength	*	479	-	1.61e3	MPa
Elongation		8	-	24	% strain
Hardness - Vickers		215	-	515	HV
Fatigue strength at 10^7 cycles	*	334	-	651	MPa
Fracture toughness	*	30	-	106	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	2.4e-4	-	7.6e-4	

Thermal properties

Melting point		1.41e3	-	1.52e3	°C
Maximum service temperature	*	415	-	530	°C
Minimum service temperature	*	-66.2	-	-35.2	°C
Thermal conductor or insulator?		Good conductor			
Thermal conductivity	*	35	-	52.1	W/m.°C
Specific heat capacity	*	440	-	520	J/kg.°C
Thermal expansion coefficient		11	-	13	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Good conductor			
Electrical resistivity	*	15	-	34	µohm.cm

Optical properties

Transparency	Opaque

Critical Materials Risk

High critical material risk?	No

Processability

Castability	1	-	2
Formability	3	-	4
Machinability	3	-	4
Weldability	5		
Solder/brazability	5		

Durability: water and aqueous solutions

Water (fresh)

	Acceptable
Water (salt)	Limited use
Soils, acidic (peat)	Acceptable
Soils, alkaline (clay)	Acceptable
Wine	Unacceptable

Durability: acids

Acetic acid (10%)	Limited use
Acetic acid (glacial)	Unacceptable
Citric acid (10%)	Unacceptable
Hydrochloric acid (10%)	Unacceptable
Hydrochloric acid (36%)	Unacceptable
Hydrofluoric acid (40%)	Unacceptable
Nitric acid (10%)	Unacceptable
Nitric acid (70%)	Unacceptable
Phosphoric acid (10%)	Unacceptable
Phosphoric acid (85%)	Unacceptable
Sulfuric acid (10%)	Unacceptable
Sulfuric acid (70%)	Unacceptable

Durability: alkalis

Sodium hydroxide (10%)	Acceptable
Sodium hydroxide (60%)	Acceptable

Durability: fuels, oils and solvents

Amyl acetate	Excellent
Benzene	Excellent
Carbon tetrachloride	Excellent
Chloroform	Excellent
Diesel oil	Excellent
Lubricating oil	Excellent
Paraffin oil (kerosene)	Excellent
Petrol (gasoline)	Excellent
Silicone fluids	Excellent
Toluene	Excellent
Turpentine	Excellent
Vegetable oils (general)	Excellent
White spirit	Excellent

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Limited use
Acetone	Excellent
Ethyl alcohol (ethanol)	Acceptable
Ethylene glycol	Acceptable
Formaldehyde (40%)	Unacceptable
Glycerol	Excellent
Methyl alcohol (methanol)	Acceptable

Durability: halogens and gases

Chlorine gas (dry)	Acceptable
Fluorine (gas)	Acceptable
O2 (oxygen gas)	Limited use
Sulfur dioxide (gas)	Acceptable

Durability: built environments

Industrial atmosphere	Limited use
Rural atmosphere	Acceptable
Marine atmosphere	Limited use
UV radiation (sunlight)	Excellent

Durability: flammability

Flammability	Non-flammable

Durability: thermal environments

Tolerance to cryogenic temperatures	Unacceptable
Tolerance up to 150 C (302 F)	Excellent
Tolerance up to 250 C (482 F)	Excellent
Tolerance up to 450 C (842 F)	Excellent
Tolerance up to 850 C (1562 F)	Unacceptable
Tolerance above 850 C (1562 F)	Unacceptable

Geo-economic data for principal component

Annual world production, principal component	2.23e9	tonne/yr
Reserves, principal component	1.59e11	tonne

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 2.24	-	2.46	kg/kg
Embodied energy (virgin grade production)	* 30.4	-	33.6	MJ/kg
Water usage	* 47.9	-	53	l/kg

Material processing: energy

Metal casting, embodied energy	* 47.9	-	54.9	MJ/kg
Metal extrusion, hot, embodied energy	* 18.8	-	25.9	MJ/kg
Metal rolling and forging, embodied energy	* 9.91	-	13.5	MJ/kg
Wire drawing, embodied energy	* 4.38	-	4.83	MJ/kg
Metal powder forming, embodied energy	* 67	-	83.3	MJ/kg
Chemical vapor deposition (CVD), embodied energy	* 9.82e3	-	1.15e4	MJ/kg
Machining, coarse, embodied energy	* 17.3	-	19.8	MJ/kg
Machining, fine, embodied energy	* 22.7	-	27.3	MJ/kg
Machining, grinding, embodied energy	* 28.1	-	34.8	MJ/kg
Machining, laser, CO2, embodied energy	* 1.17	-	1.36	MJ/m

Material processing: climate change

Metal casting, climate change (CO2-eq)	* 3.45	-	3.98	kg/kg
Metal extrusion, hot, climate change (CO2-eq)	* 1.23	-	1.7	kg/kg
Metal rolling and forging, climate change (CO2-eq)	* 0.648	-	0.886	kg/kg
Wire drawing, climate change (CO2-eq)	* 0.286	-	0.317	kg/kg
Metal powder forming, climate change (CO2-eq)	* 5.47	-	6.8	kg/kg
Chemical vapor deposition (CVD), climate change (CO2-eq)	* 808	-	943	kg/kg
Machining, coarse, climate change (CO2-eq)	* 1.13	-	1.3	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.48	-	1.79	kg/kg
Machining, grinding, climate change (CO2-eq)	* 1.84	-	2.28	kg/kg
Machining, laser, CO2, climate change (CO2-eq)	* 0.0962	-	0.112	kg/m

Recycling and end of life

Recycle	✓			
Functional recycle	✓			
Climate change (CO2-eq), recycling	* 0.605	-	0.664	kg/kg
Embodied energy, recycling	* 9.42	-	10.4	MJ/kg
Recycle fraction in current supply	40			%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Environmental notes

Steels are not particularly energy intensive to make, and are easily and widely recycled.

Supporting information

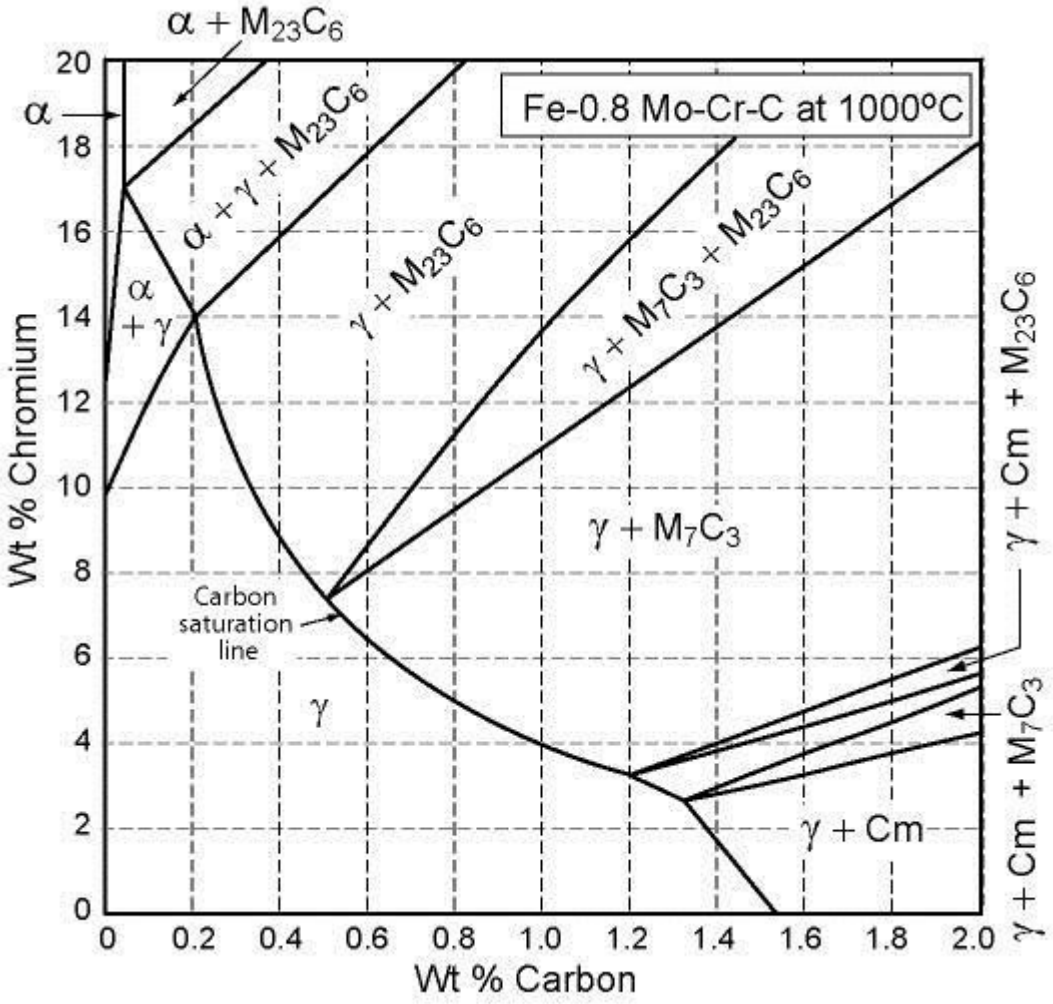
Design guidelines

Low alloy steels are heat treatable - most other carbon steels are not - and so are used for applications where hardness or strength is an important feature, particularly in large sections. They have greater abrasion resistance, higher toughness and better strength at high temperatures than plain carbon steels. Alloy steels with carbon content of 0.30 to 0.37 % are used for moderate strength and great toughness; 0.40 - 0.42% for higher strength and good toughness; 0.45 - 0.50% for high hardness and strength with moderate toughness; 0.50-0.62% for hardness (springs and tools); 1% for high hardness and abrasion resistance (ball bearings or rollers).

Technical notes

The SAE-AISI system for low alloy steels works the same way as that for plain carbon steels. Each steel has a four-digit code; the first two digits indicate the major alloying elements, the second two give the amount of carbon, in hundredths of a percent. Typical are the nickel-chrome-molybdenum steels with the designation 43xx, but the alloying elements can include any of the following: more than 2% silicon, more than 0.4% copper, more than 0.1% molybdenum, more than 0.5% nickel, more than 0.5% chromium.

Phase diagram



Phase diagram description

High strength low alloy steels contain small (<5%) additions of chromium (Cr), molybdenum (Mo), nickel (Ni) and carbon (C).

Typical uses

Springs, tools, ball bearings, rollers, crankshafts, gears, connecting rods, knives and scissors, pressure vessels.

Links

ProcessUniverse

Producers

Reference